Robust Model Predictive Control for Coupled Subsystems as a Convex Semi-Infinite Program

Abhinav Raghuvanshi

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Original Problem

Consider M coupled subsystems indexed by

$$i \in \mathcal{M} = \{1, \dots, M\}.$$

Each subsystem has a neighbor set $\mathcal{N}_i$. The local dynamics are

$$x_i(k+1) = A_{ii}x_i(k) + B_{ii}u_i(k) + \sum_{j \in \mathcal{N}_i} A_{ij}x_j(k) + w_i(k), \qquad w_i(k) \in \mathbb{W}_i.$$

A tube decomposition is used:

$$x_i = \bar{x}_i + e_i,$$

together with the feedback control law

$$u_i = \bar{u}_i + K_i e_i.$$

The local feedback gain K_i is chosen such that

$$A_{ii} + B_{ii}K_i$$

is Schur. Let $\mathcal{E}_i$ be a robust positively invariant set for the local error dynamics. The nominal state and input constraints are tightened as

$$\bar{\mathcal{X}}_i = \mathcal{X}_i \ominus \mathcal{E}_i, \qquad \bar{\mathcal{U}}_i = \mathcal{U}_i \ominus K_i\mathcal{E}_i.$$

At subsystem i, at time k, the distributed robust MPC problem is

$$\begin{aligned}
\min_{\bar{\mathbf{u}}_i, \bar{\mathbf{x}}_i} \quad & \sum_{l=0}^{N-1} \ell_i\big(\bar{x}_i(l \mid k), \bar{u}_i(l \mid k)\big) \\
\text{s.t.} \quad & \bar{x}_i(l+1 \mid k) = A_{ii}\bar{x}_i(l \mid k) + B_{ii}\bar{u}_i(l \mid k) + \sum_{j \in \mathcal{N}_i} A_{ij}\hat{x}_j(l \mid k), \\
& \bar{x}_i(l \mid k) \in \bar{\mathcal{X}}_i, \\
& \bar{u}_i(l \mid k) \in \bar{\mathcal{U}}_i, \\
& x_i(k) - \bar{x}_i(0 \mid k) \in \mathcal{E}_i, \\
& N \geq N^\star, \\
& l = 0, \dots, N-1.
\end{aligned}$$

The stage cost is quadratic:

$$\ell_i(\bar{x}_i, \bar{u}_i) = \bar{x}_i^\top Q_i \bar{x}_i + \bar{u}_i^\top R_i \bar{u}_i, \qquad Q_i \succ 0, \quad R_i \succ 0.$$

The trajectories

$$\hat{x}_j(l \mid k), \qquad j \in \mathcal{N}_i,$$

are the assumed neighbor trajectories broadcast over the communication graph at time k.

After solving the local optimization problem, subsystem i applies the receding-horizon input

$$u_i(k) = \bar{u}_i^\star(0 \mid k) + K_i\big(x_i(k) - \bar{x}_i^\star(0 \mid k)\big).$$

The condition

$$N \geq N^\star$$

replaces the usual terminal cost and terminal set.

Convex Semi-Infinite Reformulation

Decision Variables

The decision variables are the nominal input sequence

$$\bar{\mathbf{u}}_i = \big(\bar{u}_i(0 \mid k), \ldots, \bar{u}_i(N-1 \mid k)\big),$$

and the nonnegative slack vector

$$\boldsymbol{\epsilon}_i = \big(\epsilon_i(0), \ldots, \epsilon_i(N-1)\big) \in \mathbb{R}^N_{\geq 0}.$$

Drop the tube decomposition and predict the actual state directly under uncertainty. The applied input becomes

$$u_i(k) = \bar{u}_i^\star(0 \mid k).$$

Predicted State as an Affine Map

Given the measured state $x_i(k)$ and assumed neighbor trajectories $\hat{x}_j(\cdot \mid k)$, define

$$\phi_i\big(l; \bar{\mathbf{u}}_i, \mathbf{w}_i\big) = A_{ii}^l x_i(k) + \sum_{m=0}^{l-1} A_{ii}^{l-1-m} \left[B_{ii}\bar{u}_i(m \mid k) + \sum_{j \in \mathcal{N}_i} A_{ij}\hat{x}_j(m \mid k) + w_i(m) \right].$$

Here,

$$\mathbf{w}_i = \big(w_i(0), \ldots, w_i(N-1)\big)$$

is the disturbance sequence over the prediction horizon. For every fixed disturbance sequence $\mathbf{w}_i$, the map

$$\bar{\mathbf{u}}_i \mapsto \phi_i\big(l; \bar{\mathbf{u}}_i, \mathbf{w}_i\big)$$

is affine.

Uncertainty Set

The uncertainty set over the horizon is

$$\mathcal{W}_i := \mathbb{W}_i^N = \left\{ \mathbf{w}_i = \big(w_i(0), \ldots, w_i(N-1)\big) : w_i(m) \in \mathbb{W}_i \right\},$$

where

$$\mathbb{W}_i$$

is compact and convex.

The Convex Semi-Infinite Program

The robust MPC problem can be written as the following convex semi-infinite program:

$$\boxed{\begin{aligned} \min_{\bar{\mathbf{u}}_i, \boldsymbol{\epsilon}_i} \quad & \sum_{l=0}^{N-1} \epsilon_i(l) \\ \text{s.t.} \quad & \ell_i\left(\phi_i\big(l; \bar{\mathbf{u}}_i, \mathbf{w}_i\big), \bar{u}_i(l \mid k)\right) \le \epsilon_i(l), \\ & \phi_i\big(l; \bar{\mathbf{u}}_i, \mathbf{w}_i\big) \in \mathcal{X}_i, \\ & \bar{u}_i(l \mid k) \in \mathcal{U}_i, \\ & \epsilon_i(l) \ge 0, \\ & \forall \mathbf{w}_i \in \mathcal{W}_i, \qquad l = 0, \dots, N-1. \end{aligned}}$$

The stage cost is

$$\ell_i(x, u) = x^\top Q_i x + u^\top R_i u,$$

where

$$Q_i \succeq 0, \qquad R_i \succ 0.$$

It is assumed that the state constraint set, input constraint set, and disturbance set satisfy

$$\mathcal{X}_i \text{ is convex}, \qquad \mathcal{U}_i \text{ is convex}, \qquad \mathbb{W}_i \text{ is compact and convex}.$$

Solution Method: gradOL

We solve the CSIP above with gradOL (Raghuvanshi, Baranwal, Chatterjee, arXiv:2601.06434). The idea is to replace the infinitely many constraints by a finite number of worst-case disturbance sequences, and to find these sequences by gradient ascent.

CSIP in Standard Form

Stack the decision variables as

$$y_i = (\bar{\mathbf{u}}_i, \boldsymbol{\epsilon}_i) \in \mathbb{R}^d, \qquad d = N(m_i + 1).$$

Let $\mathcal{X}_i = \{x : \ h_i(x) \le 0\}$ with h_i convex, and define, for $l = 0, \dots, N-1$,

$$g_l^{\mathrm{c}}(y_i, \mathbf{w}_i) = \ell_i\big(\phi_i(l; \bar{\mathbf{u}}_i, \mathbf{w}_i), \bar{u}_i(l \mid k)\big) - \epsilon_i(l), \qquad g_l^{\mathrm{x}}(y_i, \mathbf{w}_i) = h_i\big(\phi_i(l; \bar{\mathbf{u}}_i, \mathbf{w}_i)\big),$$

$$f_i(y_i, \mathbf{w}_i) = \max_l \ \max\big\{g_l^{\mathrm{c}}(y_i, \mathbf{w}_i),\, g_l^{\mathrm{x}}(y_i, \mathbf{w}_i)\big\}.$$

The constraints that do not depend on $\mathbf{w}_i$ form the set

$$\mathcal{Y}_i = \big\{y_i : \ \bar{u}_i(l \mid k) \in \mathcal{U}_i, \ \epsilon_i(l) \ge 0, \ l = 0, \dots, N-1\big\}.$$

The CSIP is then

$$\min_{y_i \in \mathcal{Y}_i} \ \mathbf{1}^\top \boldsymbol{\epsilon}_i \quad \text{s.t.} \quad f_i(y_i, \mathbf{w}_i) \le 0 \quad \forall\, \mathbf{w}_i \in \mathcal{W}_i.$$

Since ϕ_i is affine in $\bar{\mathbf{u}}_i$, $f_i(\cdot, \mathbf{w}_i)$ is convex for every $\mathbf{w}_i$.

Finite Max-Min Reformulation

Take d disturbance sequences $\overline{\mathbf{w}} = (\mathbf{w}_i^{(1)}, \ldots, \mathbf{w}_i^{(d)}) \in \mathcal{W}_i^d$ and define the value function

$$\mathcal{G}_i(\overline{\mathbf{w}}) = \min_{y_i \in \mathcal{Y}_i} \Big\{ \mathbf{1}^\top \boldsymbol{\epsilon}_i \ : \ f_i(y_i, \mathbf{w}_i^{(r)}) \leq 0, \ r = 1, \ldots, d \Big\}.$$

Assume $\mathcal{Y}_i$ is closed and convex with nonempty interior, $\mathcal{W}_i$ is compact, and the robust feasible set has nonempty interior (Slater). Then (Das et al., 2022, Thm. 1)

$$\boxed{\text{value of the CSIP} = \max_{\overline{\mathbf{w}} \in \mathcal{W}_i^d} \mathcal{G}_i(\overline{\mathbf{w}}).}$$

Since $R_i \succ 0$, the robust objective is strictly convex in $\bar{\mathbf{u}}_i$, so the CSIP has a unique solution. Hence the minimizer of the inner problem at a maximizer $\overline{\mathbf{w}}^\star$ solves the CSIP. Also, for any $\overline{\mathbf{w}}$, $\mathcal{G}_i(\overline{\mathbf{w}})$ is a lower bound on the CSIP value.

Algorithm

Inner step (log-barrier). For fixed $\overline{\mathbf{w}}$ and barrier parameter $\alpha > 0$, solve

$$y_i^\star(\overline{\mathbf{w}}) \in \arg\min_{y_i} \ \alpha \, \mathbf{1}^\top \boldsymbol{\epsilon}_i - \sum_{r=1}^{d} \sum_{l=0}^{N-1} \Big[\log\big(-g_l^{\mathrm{c}}(y_i, \mathbf{w}_i^{(r)})\big) + \log\big(-g_l^{\mathrm{x}}(y_i, \mathbf{w}_i^{(r)})\big) \Big] - B_{\mathcal{Y}_i}(y_i),$$

where $B_{\mathcal{Y}_i}$ is a log-barrier for $\mathcal{Y}_i$. For box sets, each linear row gets its own log term. Then $\mathcal{G}_i(\overline{\mathbf{w}}) = \mathbf{1}^\top \boldsymbol{\epsilon}_i^\star(\overline{\mathbf{w}})$.

Outer step (projected gradient ascent). The gradient $\nabla_{\overline{\mathbf{w}}} \mathcal{G}_i$ is computed by automatic differentiation through the inner solver, and

$$\overline{\mathbf{w}}^{\,t+1} = \Pi_{\mathcal{W}_i^d} \Big(\overline{\mathbf{w}}^{\,t} + \eta \, \nabla_{\overline{\mathbf{w}}} \mathcal{G}_i(\overline{\mathbf{w}}^{\,t}) \Big).$$

For a box $\mathbb{W}_i$, the projection is elementwise clipping.

1. Initialize $\overline{\mathbf{w}}^{\,0} \in \mathcal{W}_i^d$; choose α, η, tolerance δ, max epochs T.
2. For $t = 0, \ldots, T-1$: solve the inner step, compute $\nabla_{\overline{\mathbf{w}}} \mathcal{G}_i$, update $\overline{\mathbf{w}}^{\,t+1}$.
3. Stop if $\|\overline{\mathbf{w}}^{\,t+1} - \overline{\mathbf{w}}^{\,t}\| < \delta$.
4. Return $y_i^\star(\overline{\mathbf{w}}^{\,t})$ and apply $u_i(k) = \bar{u}_i^\star(0 \mid k)$.

The outer problem is in general non-concave and non-smooth, so gradOL is guaranteed only to reach a stationary point of $\mathcal{G}_i$, not necessarily a global maximizer.

1 Experiment

Experiment Details

For the numerical experiment, consider a single local subsystem with state dimension $n_i = 2$ and input dimension $m_i = 1$. The prediction horizon is chosen as

$$N = 4.$$

The system matrices are

$$A_{ii} = \begin{bmatrix} 1 & 0.1 \\ 0 & 1 \end{bmatrix}, \qquad B_{ii} = \begin{bmatrix} 0 \\ 0.1 \end{bmatrix},$$

with initial condition

$$x_i(k) = \begin{bmatrix} 0.6 \\ -0.3 \end{bmatrix}.$$

The quadratic stage cost uses

$$Q_i = I_2, \qquad R_i = 0.1.$$

The disturbance, state, and input bounds are chosen as

$$|w_i(l)| \le 0.05, \qquad |x_i(l)| \le 3, \qquad |\bar{u}_i(l)| \le 2.$$

The decision variable is $z = (\bar{\mathbf{u}}_i, \boldsymbol{\epsilon}_i)$ and an outer maximization happens over sampled disturbance sequences. The outer maximization is run for at most 40 epochs. The plots below show the evolution of the SIP objective, the optimized slack variables, the resulting input sequence, and the state trajectories under nominal and disturbed rollouts.

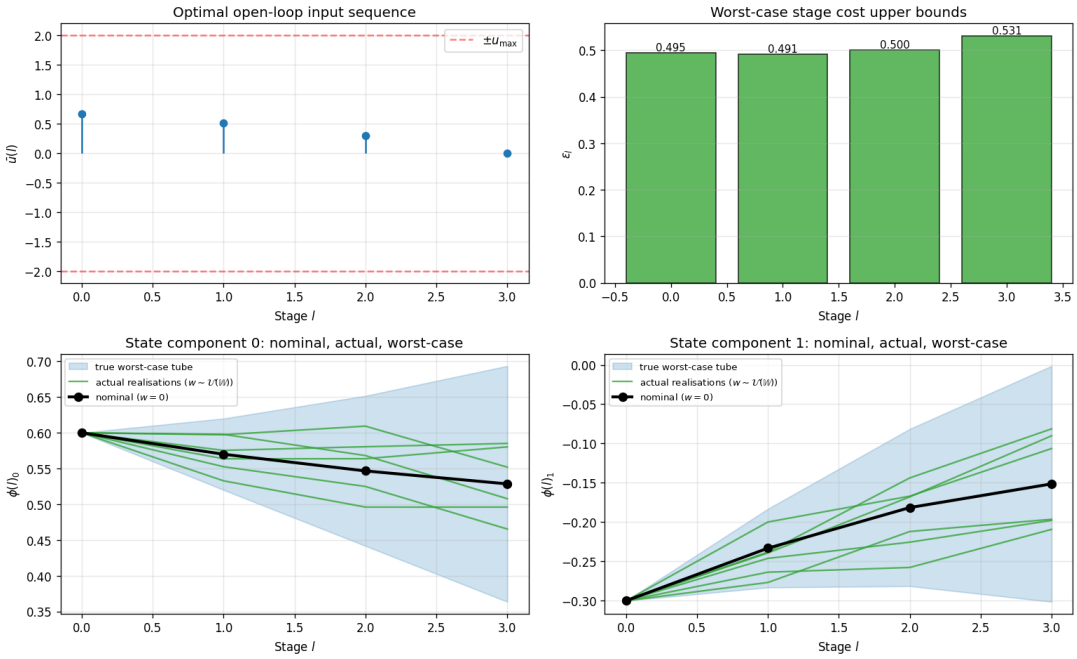


Figure 1: Trajectories

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epoch   1 | f(z) = 1.55737 | ||u|| = 0.0727 | ||grad_u|| = 4.282e+00
epoch   2 | f(z) = 1.65147 | ||u|| = 0.0774 | ||grad_u|| = 4.496e+00
epoch   3 | f(z) = 1.75487 | ||u|| = 0.0877 | ||grad_u|| = 4.701e+00
epoch   4 | f(z) = 1.85232 | ||u|| = 0.1007 | ||grad_u|| = 4.879e+00
epoch   5 | f(z) = 1.90240 | ||u|| = 0.1091 | ||grad_u|| = 4.983e+00
epoch   6 | f(z) = 1.93538 | ||u|| = 0.1159 | ||grad_u|| = 5.058e+00
epoch   7 | f(z) = 1.95597 | ||u|| = 0.1212 | ||grad_u|| = 5.065e+00
epoch   8 | f(z) = 1.97703 | ||u|| = 0.1271 | ||grad_u|| = 5.122e+00
epoch   9 | f(z) = 1.99068 | ||u|| = 0.1311 | ||grad_u|| = 5.128e+00
epoch  10 | f(z) = 2.00076 | ||u|| = 0.1344 | ||grad_u|| = 5.120e+00
epoch  11 | f(z) = 2.00420 | ||u|| = 0.1356 | ||grad_u|| = 5.149e+00
epoch  12 | f(z) = 2.00582 | ||u|| = 0.1362 | ||grad_u|| = 5.127e+00
epoch  13 | f(z) = 2.00745 | ||u|| = 0.1369 | ||grad_u|| = 5.156e+00
epoch  14 | f(z) = 2.00911 | ||u|| = 0.1376 | ||grad_u|| = 5.145e+00
epoch  15 | f(z) = 2.01080 | ||u|| = 0.1384 | ||grad_u|| = 5.134e+00
epoch  16 | f(z) = 2.01251 | ||u|| = 0.1392 | ||grad_u|| = 5.146e+00
epoch  17 | f(z) = 2.01426 | ||u|| = 0.1402 | ||grad_u|| = 5.165e+00
epoch  18 | f(z) = 2.01605 | ||u|| = 0.1412 | ||grad_u|| = 5.141e+00
epoch  19 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  20 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  21 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  22 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  23 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  24 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  25 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  26 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  27 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  28 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  29 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.159e+00
epoch  30 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  31 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  32 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  33 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  34 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  35 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  36 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  37 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  38 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  39 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00
epoch  40 | f(z) = 2.01716 | ||u|| = 0.1418 | ||grad_u|| = 5.170e+00

Final optimal bar_u = [0.6690005  0.5164179  0.30132556 0.        ]
Final stage slacks  = [0.49476624 0.49122432 0.50007445 0.531098  ]
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Figure 2: gradOL epochs

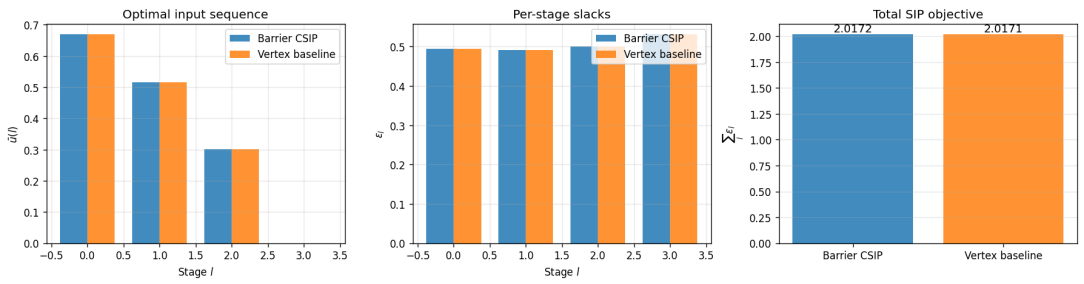


Figure 3: Comparison with baseline algorithm